AS · Unit 1 · WCH11 Structure, Bonding and Introduction to Organic Chemistry

# Alkenes

Alkenes contain a carbon–carbon double bond, and that one bond changes everything. It makes alkenes far more reactive than alkanes, it stops the molecule twisting (which creates a new kind of isomer), and it lets small molecules join into the long chains of addition polymers. This topic explains all three, starting from what a double bond actually is.

## SPEC 5.1 Alkenes: unsaturated hydrocarbons with σ and π bonds

**WHAT THIS SPEC POINT ASKS**

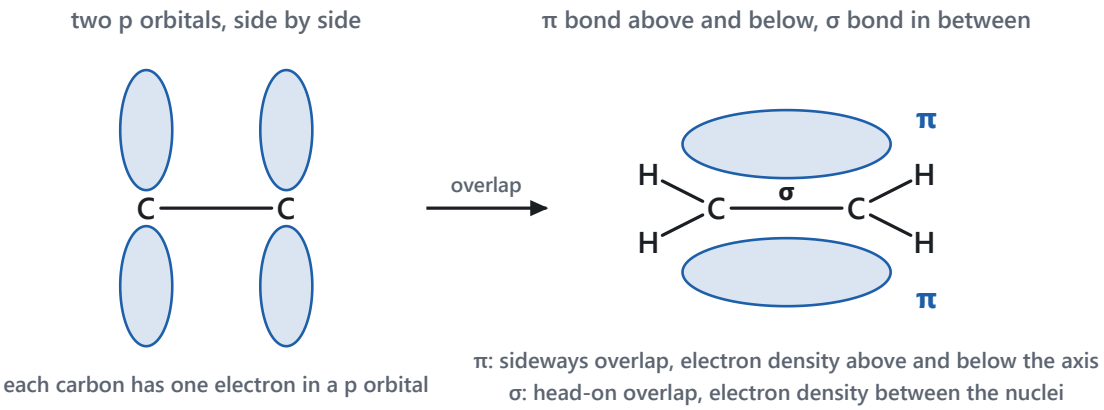
Know the general formula of alkenes. Understand that alkenes and cycloalkenes are unsaturated hydrocarbons, with a C=C double bond made of one σ bond and one π bond.

### Key facts

- **Alkenes**: hydrocarbons with **one C=C double bond**. General formula $\mathbf{C_nH_{2n}}$. Ethene $C_2H_4$, propene $C_3H_6$, butene $C_4H_8$.
- **Cycloalkenes** (a ring plus one C=C): $\mathbf{C_nH_{2n-2}}$. Cyclohexene $C_6H_{10}$.
- **Unsaturated**: the molecule contains a C=C double bond, so more atoms can be **added** across it.
- Around each carbon of the C=C there are three bonding regions, so the shape is **trigonal planar**, bond angle **about 120°**. The two carbons and the four atoms attached to them all lie in **one plane**.

### What the double bond is made of

- **σ (sigma) bond**: formed by **head-on (end-on) overlap** of two orbitals, directly between the two nuclei. The electron density sits **on the line between the nuclei**. Every single bond is a σ bond.
- **π (pi) bond**: formed by **sideways overlap of two p orbitals**, one on each carbon. The electron density sits **above and below** the plane of the molecule, not between the nuclei.
- A C=C double bond = **one σ bond + one π bond**.




Each carbon's p orbital (left) overlaps sideways with its neighbour's to make the π bond (right): two regions of electron density, above and below the σ bond.

## Why the π bond makes alkenes reactive

- The π electrons are further from the nuclei and **exposed** above and below the molecule, so they are easier to reach.
- The π bond is **weaker** than the σ bond. C=C (612 kJ $mol^{-1}$) is **less** than twice C–C (2 × 347 = 694 kJ $mol^{-1}$).
- The C=C is a region of **high electron density**, so it attracts **electrophiles** (5.6). In addition reactions the π bond breaks and the σ bond stays.

WORKED EXAMPLE

### Worked example: counting σ and π bonds

Propene, $CH_2{=}CHCH_3$.

- 6 C–H bonds: 6 σ
- 1 C–C bond: 1 σ
- 1 C=C bond: 1 σ + 1 π

Total: **8 σ and 1 π**.

Rule: every single bond is 1 σ; every double bond is 1 σ + 1 π.

MARK-SCHEME WORDING

- Drawing the π bond (Jan 2024): **two** regions of electron density (lobes), one **above** and one **below** the σ bond, and the σ bond shown between the nuclei.
- σ = "end-on / head-on overlap"; π = "**sideways** overlap of **p** orbitals".

Often asked as: describe or draw how a π bond forms; count σ and π bonds in an unfamiliar molecule (MCQ, Jan 2026); give a general formula (cycloalkene, Oct 2024).

**COMMON MISTAKE**

Drawing the π bond in line with the σ bond, or as one lobe only (WCH11 Jan 2022 ER).

Drawing a single line for the σ bond with no indication of overlapping orbitals when the question asks for orbitals (WCH11 Jan 2024 ER).

Saying the π bond is "stronger" because a double bond is stronger. The **double bond** is stronger than a single bond, but the **π part** of it is weaker than the σ part.

# SPEC 5.2 Geometric (E/Z) isomerism

**WHAT THIS SPEC POINT ASKS**

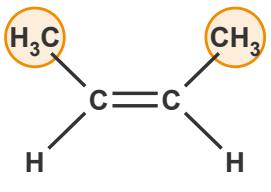
Explain geometric isomerism using restricted rotation about C=C and the nature of the groups on each carbon of the double bond.

## Stereoisomers

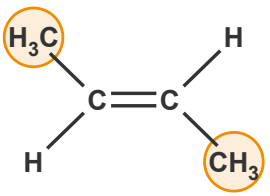
**Stereoisomers** have the **same structural formula** (atoms joined in the same order) but a **different arrangement of atoms in space**. Geometric (E/Z or cis/trans) isomerism is one kind.

## Two conditions, both needed

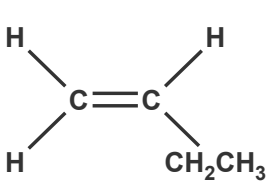
1. **Restricted rotation about the C=C.** Rotating one end of a double bond would twist the p orbitals apart and **break the π bond**. That needs a lot of energy, so it doesn't happen at room temperature. (Around a C–C single bond, rotation is free, so alkanes don't form geometric isomers.)
2. **Two different groups on each carbon of the C=C.** If either carbon carries two identical groups, swapping them changes nothing.



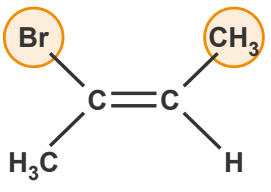

**Z-but-2-ene**
(cis-but-2-ene)




**E-but-2-ene**
(trans-but-2-ene)




**but-1-ene**
two H on one carbon: no E/Z




**Z-2-bromobut-2-ene**
Br and $CH_3$ on the same side

The higher-priority group on each carbon is circled. In but-1-ene the left-hand carbon carries two hydrogen atoms, so swapping them changes nothing and there are no E/Z isomers.

## Check each C=C carbon

| Alkene | Groups on carbon 1 of C=C | Groups on carbon 2 of C=C | Geometric isomers? |
|---|---|---|---|
| propene $CH_2=CHCH_3$ | H, H | H, $CH_3$ | **no** (two H on one carbon) |
| but-1-ene $CH_2=CHCH_2CH_3$ | H, H | H, $C_2H_5$ | **no** |
| but-2-ene $CH_3CH=CHCH_3$ | H, $CH_3$ | H, $CH_3$ | **yes** |
| 2-methylpropene $(CH_3)_2C=CH_2$ | $CH_3$, $CH_3$ | H, H | **no** |
| 2-methylbut-2-ene $(CH_3)_2C=CHCH_3$ | $CH_3$, $CH_3$ | H, $CH_3$ | **no** |
| pent-2-ene $CH_3CH=CHCH_2CH_3$ | H, $CH_3$ | H, $C_2H_5$ | **yes** |
| 1,2-dichloroethene CHCl=CHCl | H, Cl | H, Cl | **yes** |

**MARK-SCHEME WORDING**

- Why isomers exist (Jan 2025, Oct 2025): "**restricted rotation** around the C=C" **and** "**two different groups / atoms** on each carbon of the double bond".
- Why no isomers (Oct 2020, isoprene): "each double bond has **two hydrogen atoms on one carbon**"; ALLOW "two identical groups / atoms attached to one carbon".
- The π bond prevents rotation; rotation would need the π bond to break.

Often asked as: explain why a named alkene has geometric isomers (2 marks); explain why another does not; draw both isomers.

**COMMON MISTAKE**

"Similar groups" instead of **different** groups; "molecules" instead of **groups** or **atoms** (WCH11 Oct 2020, Jan 2024 ER).

"The double bond restricts rotation **of the molecule**" or "**allows** restricted rotation". Say: there is no rotation **about the C=C bond** (WCH11 Oct 2020, Jan 2024 ER).

Checking only one carbon of the C=C. Check both.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Molecules with **several C=C bonds**: test each double bond separately. Only C=C bonds with two different groups on both carbons give E/Z isomers. A molecule whose other C=C bonds each carry two identical groups has just two geometric isomers (WCH11 Jan 2021, α-farnesene; WCH11 Jan 2022, ocimene).

# SPEC 5.3 The E–Z naming system

**WHAT THIS SPEC POINT ASKS**

Understand E–Z naming of geometric isomers, and why it is needed when cis/trans naming breaks down.

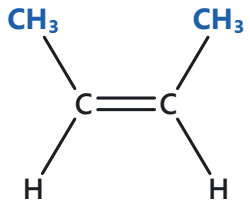
## Cis/trans and its limit

- **cis**: two identical groups on the **same side** of the C=C. **trans**: on **opposite sides**.
- This only works when the two carbons share a group to compare (usually H or $CH_3$). If all four groups are different (e.g. $CBrCl{=}CHCH_3$), there is nothing to compare, so cis/trans can't be used. E/Z works for every case.

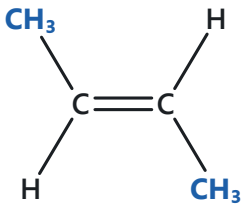
## E/Z: the priority rules (Cahn–Ingold–Prelog)

1. Look at **each carbon of the C=C separately**.
2. Compare the two **atoms directly attached** to that carbon. The one with the **higher atomic number** has **higher priority**. (Br 35 > Cl 17 > O 8 > C 6 > H 1.)
3. If the two attached atoms are the same (e.g. both C), list the atoms attached **to each of them** in order of atomic number and compare at the **first point of difference**.
4. **Z** (*zusammen*, together): the two higher-priority groups are on the **same side**.
   **E** (*entgegen*, opposite): the two higher-priority groups are on **opposite sides**.

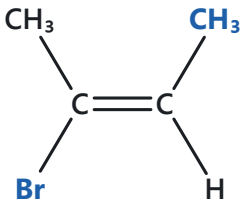
Memory aid: **Z** = "**Z**ame **Z**ide".



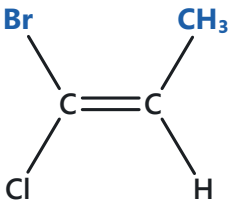

**(Z)-but-2-ene**
cis: $CH_3$ groups on the same side




**(E)-but-2-ene**
trans: $CH_3$ groups on opposite sides




**(E)-2-bromobut-2-ene**
$CH_3$ groups 'cis', but it is E




**(Z)-1-bromo-1-chloroprop-1-ene**
no shared group: cis/trans fails

Higher-priority group on each carbon in blue. For but-2-ene, Z = cis and E = trans. In 2-bromobut-2-ene the methyls are "cis" but Br outranks $CH_3$, so the isomer is E: the two systems don't always agree.

## Is this 2-bromobut-2-ene E or Z?

1. Look at the left-hand carbon of the C=C. It carries Br and $CH_3$. The atoms bonded directly to the carbon are Br (atomic number 35) and C (atomic number 6). **Br has priority.**
2. Look at the right-hand carbon. It carries $CH_3$ and H: C (6) beats H (1), so **$CH_3$ has priority.**
3. Both priority groups, Br and the right-hand $CH_3$, are above the double bond: same side, so this is **Z-2-bromobut-2-ene**.

*Notice that the two $CH_3$ groups are on opposite sides, so anyone using cis/trans by comparing the methyl groups would call it "trans". That clash is why the E/Z system, which works for any groups, is the one to use.*

## Drawing E and Z isomers: the method

1. Draw the C=C horizontally.
2. Put the two groups on each carbon at about 120°: one up and one down on the left carbon, one up and one down on the right carbon.
3. Mark the higher-priority group on each carbon.
4. Both higher-priority groups up (or both down) → **Z**. One up, one down → **E**.
5. To draw the other isomer, **swap the two groups on one carbon only**. Don't redraw the chain differently, or you may change the molecule.

WORKED EXAMPLE

### Worked example: (Z)-3-methylpent-2-ene

Structure: $CH_3CH=C(CH_3)CH_2CH_3$. The C=C is between C2 and C3.

- **C2** carries $CH_3$ and H. C beats H → **$CH_3$** has priority.
- **C3** carries $CH_3$ and $CH_2CH_3$. Both attach through C: a tie. Go one atom further. $CH_3$ has (H, H, H); $CH_2CH_3$ has (**C**, H, H). C beats H at the first difference → **$CH_2CH_3$** has priority.
- **Z**: draw the $CH_3$ on C2 and the $CH_2CH_3$ on C3 on the **same side**.

Notice that in the Z isomer the two $CH_3$ groups are on **opposite** sides. Cis/trans by the methyls would give the wrong answer.

MARK-SCHEME WORDING

- Z (Jan 2021): "Z isomer has the **highest priority groups** on the **same side** of the C=C"; priority by **atomic number**, not mass.
- Trans (Jun 2024): "the two (identical) groups / alkyl groups are on **opposite sides** of the C=C". "Molecule" for "group" does not score.
- A full name has the **E or Z**, the **position of the double bond**, the stem and **-ene** (Jan 2024): (E)-pent-2-ene, not "pentene" or "E-pentene".

Often asked as: draw the Z isomer of a named alkene and explain why it is Z; name a drawn isomer; MCQ on E/Z.

COMMON MISTAKE

Ranking groups by **mass** or by size instead of the **atomic number** of the first atom (WCH11 Jan 2021 ER).

Drawing the E isomer when Z was asked, or the **same** isomer twice in a different orientation (WCH11 Jan 2019, Oct 2019 ER).

Leaving out E/Z or the double-bond position in a name; "dimethylethene" for but-2-ene (WCH11 Jan 2024 ER).

In skeletal formulae, moving where the chain attaches when you swap groups (WCH11 Jan 2022 ER).

## SPEC 5.4 Reactions of alkenes

WHAT THIS SPEC POINT ASKS

Describe these alkene reactions: (i) + hydrogen (nickel catalyst) → alkane; (ii) + halogen → dihalogenoalkane; (iii) + hydrogen halide → halogenoalkane; (iv) + steam (acid catalyst) → alcohol; (v)

oxidation by acidified potassium manganate(VII) → diol.

## The five reactions at a glance

| Reagent | Conditions | Ethene gives | Type |
|---|---|---|---|
| **$H_2$** | **nickel** catalyst, about 150 °C (Pt or Pd also work) | ethane $CH_3CH_3$ | addition / reduction (hydrogenation) |
| **$Br_2$** (or $Cl_2$) | room temperature; liquid bromine, bromine in an organic solvent, or bromine water | 1,2-dibromoethane $CH_2BrCH_2Br$ | electrophilic addition |
| **HBr** (or HCl, HI) | room temperature, gaseous HBr | bromoethane $CH_3CH_2Br$ | electrophilic addition |
| **$H_2O$** as **steam** | **phosphoric(V) acid ($H_3PO_4$)** catalyst; industrially about 300 °C and 60–70 atm | ethanol $CH_3CH_2OH$ | addition (hydration) |
| **$KMnO_4$ / $H^+$** (acidified potassium manganate(VII)) | room temperature, shake | ethane-1,2-diol $HOCH_2CH_2OH$ | oxidation |

## Equations

| Reaction | Ethene | Propene |
|---|---|---|
| Hydrogenation | $CH_2=CH_2 + H_2 \rightarrow CH_3CH_3$ | $CH_2=CHCH_3 + H_2 \rightarrow CH_3CH_2CH_3$ (propane) |
| Halogen | $CH_2=CH_2 + Br_2 \rightarrow CH_2BrCH_2Br$ | $CH_2=CHCH_3 + Br_2 \rightarrow CH_2BrCHBrCH_3$ (**1,2-dibromopropane**) |
| Hydrogen halide | $CH_2=CH_2 + HBr \rightarrow CH_3CH_2Br$ | $CH_2=CHCH_3 + HBr \rightarrow CH_3CHBrCH_3$ (**2-bromopropane**, major) + $CH_3CH_2CH_2Br$ (1-bromopropane, minor) |
| Steam | $CH_2=CH_2 + H_2O \rightarrow CH_3CH_2OH$ | $CH_2=CHCH_3 + H_2O \rightarrow CH_3CH(OH)CH_3$ (**propan-2-ol**, major) + $CH_3CH_2CH_2OH$ (propan-1-ol, minor) |
| $KMnO_4$ / $H^+$ | $CH_2=CH_2 + [O] + H_2O \rightarrow HOCH_2CH_2OH$ | $CH_2=CHCH_3 + [O] + H_2O \rightarrow HOCH_2CH(OH)CH_3$ (**propane-1,2-diol**) |

- **[O]** stands for oxygen from the oxidising agent. Use it to balance oxidation equations.
- With $KMnO_4$ you see **purple → colourless**. The two –OH groups go on the **two carbons that were joined by the C=C** (adjacent carbons).
- Hydrogenation is used to harden vegetable oils into margarine (it removes some C=C bonds).
- For an **unsymmetrical** alkene (propene) with HBr or steam, two products are possible. Which one is major is explained in 5.6.

## Thinking it through with a cyclic alkene

Cyclohexene behaves the same way. With $H_2$/Ni → cyclohexane; with $Br_2$ → 1,2-dibromocyclohexane; with HBr → bromocyclohexane; with steam/$H_3PO_4$ → cyclohexanol; with $KMnO_4/H^+$ → cyclohexane-1,2-diol. The ring stays; only the C=C changes.

WORKED EXAMPLE

### Worked example: how many C=C bonds?

0.500 g of limonene, $C_{10}H_{16}$ ($M_r$ 136.0), reacts completely with 176 cm³ of hydrogen at room temperature and pressure (molar volume 24.0 dm³ mol$^{-1}$).

n(limonene) = 0.500 ÷ 136.0 = 0.003676 mol

n($H_2$) = 0.176 ÷ 24.0 = 0.007333 mol

Ratio $H_2$ : limonene = 0.007333 ÷ 0.003676 = 1.99 ≈ **2**

One $H_2$ adds per C=C, so limonene has **2 C=C bonds**. (Its formula, $C_{10}H_{16}$, is 6 H short of $C_{10}H_{22}$: two C=C plus one ring.)

WORKED EXAMPLE

### Worked example: iodine number

The iodine number is the mass of $I_2$ (in g) that adds to 100 g of a fat or oil. One $I_2$ adds per C=C.

Triolein ($M_r$ 885.4) has 3 C=C, so 1 mol adds 3 mol $I_2$ = 3 × 253.8 = 761.4 g.

Iodine number = 761.4 × 100 ÷ 885.4 = **86.0**

Higher iodine number = more C=C = more unsaturated.

MARK-SCHEME WORDING

- Hydrogenation catalyst: "**nickel**" (ALLOW Pd, Pt, Rh). IGNORE heat. Acids or zeolite are wrong; a catalyst listed with other substances is not awarded (Oct 2025).

- Hydration (Jan 2026, 2 marks): "**steam**" (ALLOW $H_2O(g)$, or water with a temperature above 100 °C; DO NOT AWARD water below 100 °C) **and** "**phosphoric(V) acid / $H_3PO_4$**" (ALLOW sulfuric acid, $H^+$). A wrong extra reagent such as $KMnO_4$ caps the answer at 1. Ignore pressure.
- $KMnO_4$: "**acidified potassium manganate(VII)**"; product a **diol with OH on adjacent carbons**.

Often asked as: give reagents and conditions; name or draw the product from propene, a branched alkene or cyclohexene; calculate the number of C=C bonds from a volume of $H_2$ or a mass of $I_2$/$Br_2$.

**COMMON MISTAKE**

Writing "phosphoric acid" without steam, or "water" without saying steam; giving "heat" or a temperature above 400 °C for hydration (WCH11 Jan 2020, Jan 2023 ER).

"$K_2MnO_4$" for potassium manganate(VII). It is **$KMnO_4$**; write the name if you're unsure (WCH11 Oct 2019 ER).

Putting both Br atoms or both OH groups on the **same** carbon; writing an alcohol instead of a diol; a 1,3-diol (WCH11 Jan 2022, Jan 2023 ER). Never bond Mn into the organic product.

"Butanol" with no position number (WCH11 Oct 2021 ER).

Forgetting to multiply by the number of C=C bonds in a moles calculation (WCH11 Jun 2022 ER).

**⚠ Beyond the spec · not part of the Edexcel IAL specification**

Examined in past papers even though the spec doesn't list it.

Hydrogenation is **not** electrophilic addition (WCH11 Jan 2020).

Hydration of two different alkenes can give a set of positional isomers: pent-1-ene and pent-2-ene between them give pentan-1-ol, pentan-2-ol and pentan-3-ol (WCH11 Oct 2025).

Iodine-number and $H_2$-volume calculations to find the number of C=C bonds (WCH11 Jan 2026).

## SPEC 5.5 Testing for C=C

**WHAT THIS SPEC POINT ASKS**

Know the test for a C=C double bond using bromine or bromine water.

## The test

| | Alkene (C=C present) | Alkane or cycloalkane (no C=C) |
|---|---|---|
| Shake with **bromine water** at room temperature | **orange → colourless** | **no change** (stays orange) |
| Shake with **acidified $KMnO_4$** | **purple → colourless** | no change (stays purple) |

- The bromine **adds across the C=C** (5.4), so the coloured $Br_2$ is used up.
- Alkanes don't react with bromine in the dark: their substitution reaction needs **UV light** (4.17).
- Do it **without heating** and **in the dark** (or at least not in strong sunlight), so a positive result can only come from addition.
- Gases such as propene can be **bubbled through** bromine water or shaken in a stoppered tube with it.

CORE PRACTICAL

### Suggested practical: comparing an alkane and an alkene

Use cyclohexane (an alkane) and cyclohexene (an alkene), a few drops of each.

1. **Bromine water**: shake. Cyclohexene decolourises it (orange → colourless). Cyclohexane: no change.
2. **Acidified $KMnO_4$**: shake. Cyclohexene: purple → colourless. Cyclohexane: stays purple.
3. **Combustion** (a few drops on a watch glass, in a fume cupboard): both burn. Cyclohexene gives a **smokier (sootier) flame** because it has a higher ratio of carbon to hydrogen.

Other suggested practicals use the same test: **cyclohexene** made from cyclohexanol (heated with concentrated phosphoric(V) acid, then distilled off) and **limonene** extracted from orange peel by steam distillation both decolourise bromine water.

**Safety:** both liquids are flammable (no flames nearby except the controlled combustion test); bromine water is harmful.

MARK-SCHEME WORDING

- Bromine water test (Jun 2019): reagent "bromine water / bromine solution / $Br_2$(aq)"; observation "**yellow / orange / (red-)brown → colourless**". ALLOW "decolourises". IGNORE heat; adding an unnecessary condition such as UV **negated** the mark.
- Write "**colourless**", not "clear". Clear means see-through, and coloured solutions are clear too.
- A ring compound with no C=C: "**no change / stays orange**" (Oct 2025).
- Liquid bromine itself is **red-brown** (Oct 2020).

Often asked as: a test to distinguish propane from propene, with both observations; a test to show a compound from a reaction scheme is unsaturated.

**COMMON MISTAKE**

"Bromine water goes clear" or "turns red". It is orange to **colourless** (WCH11 Oct 2019, Jan 2020, Jan 2021 ER).

Writing about two layers when the substances tested were **gases** (propane, propene) (WCH11 Oct 2019 ER).

Describing the product ("1,2-dibromopropane forms") when the question asks what you **see** (WCH11 Jan 2021 ER).

## SPEC 5.6 Electrophilic addition: mechanisms

**WHAT THIS SPEC POINT ASKS**

Describe, with curly-arrow diagrams and evidence, the electrophilic addition of (i) bromine and hydrogen bromide to ethene and (ii) hydrogen bromide to propene. Arrows start from a bond or a lone pair. Know the relative stability of primary, secondary and tertiary carbocations.

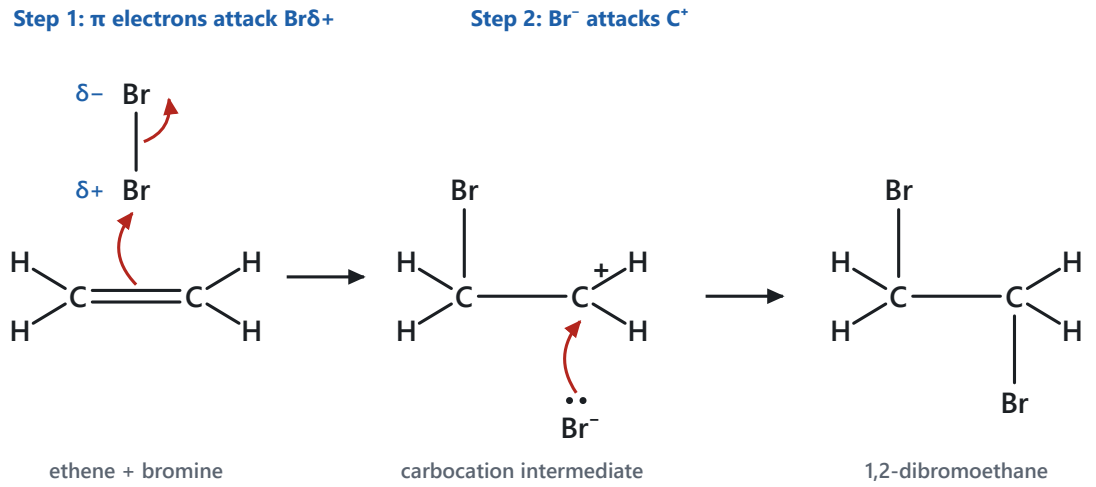
### Why "electrophilic addition"?

- The C=C is a region of **high electron density** (the π bond).
- An **electrophile** (electron-pair acceptor, 4.8) is attracted to it.
- The π electrons form a new bond to the electrophile, and two new single bonds end up where the double bond was: an **addition**.

### Rules for every curly arrow

- A full curly arrow = movement of a **pair** of electrons.
- It **starts** at a **bond** (C=C, H–Br, Br–Br) or a **lone pair**. It **ends** at the atom that receives the electrons.
- Show **δ+ / δ−** on the electrophile, a **full charge** on the carbocation (+) and on the halide ion (−), and a **lone pair** on the halide ion.

# Mechanism 1: bromine + ethene




Bromine + ethene. Step 1: two arrows (C=C to Brδ+; Br–Br bond to Brδ−). Step 2: lone pair on $Br^-$ to $C^+$.

1. **The dipole is induced.** $Br_2$ is non-polar. As it approaches, the electron-rich C=C **repels** the electrons in the Br–Br bond, so the nearer Br becomes **δ+** and the further Br **δ−**.
2. **Arrow 1** starts in the **middle of the C=C** and points to the **Brδ+**. The π electrons form a new C–Br bond.
3. **Arrow 2** starts on the **Br–Br bond** and points to the **Brδ−**. The Br–Br bond breaks **heterolytically**, giving $Br^-$.
4. **Intermediate**: one carbon now has the Br; the other carbon has lost its share of the π electrons, has only three bonds and is a **carbocation** ($C^+$). $Br^-$ is the other product of step 1.
5. **Arrow 3** starts at a **lone pair on $Br^-$** and points to the **$C^+$**. A new C–Br bond forms.
6. **Product**: $CH_2BrCH_2Br$, 1,2-dibromoethane.

## Mechanism: ethene + bromine → 1,2-dibromoethane

1. The π bond induces a dipole in $Br_2$: $Br^{\delta+}$–$Br^{\delta-}$.
2. A curly arrow from the π bond to $Br^{\delta+}$ shows the π electron pair forming a new C–Br bond. At the same time, a curly arrow from the Br–Br bond to $Br^{\delta-}$ shows that bond breaking **heterolytically**: the far bromine leaves as a bromide ion, $Br^-$.
3. That leaves a **carbocation intermediate**: the other carbon has lost its share of the π electrons and carries a positive charge.

4. A curly arrow from a **lone pair** on $Br^-$ to $C^+$ shows the bromide ion forming the second C–Br bond. The product is 1,2-dibromoethane, $CH_2BrCH_2Br$.

*Exam answers need: the δ+ and δ− on $Br_2$, the arrow from the π bond (the middle of the C=C), the arrow from the Br–Br bond, the structure of the carbocation with its + charge, a lone pair on $Br^-$ with the final arrow starting from it.*

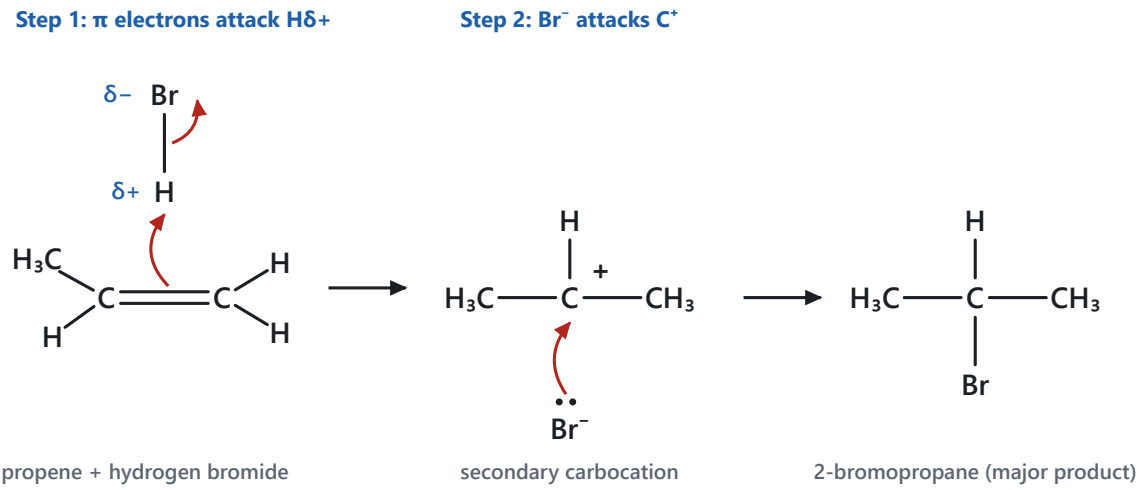
## Mechanism 2: hydrogen bromide + ethene

HBr is **already polar**: Br is more electronegative, so it is **Hδ+–Brδ−**. The pattern is the same as for $Br_2$:

1. **Arrow 1**: from the **C=C** to the **Hδ+**.
2. **Arrow 2**: from the **H–Br bond** to the **Brδ−** (heterolytic fission, giving $Br^-$).
3. Intermediate: $CH_3CH_2^+$ (a primary carbocation) and $Br^-$.
4. **Arrow 3**: from a **lone pair on $Br^-$** to the **$C^+$**.
5. Product: $CH_3CH_2Br$, bromoethane.

Ethene is symmetrical, so it doesn't matter which carbon gets the H: there is only one product.

## Mechanism 3: hydrogen bromide + propene



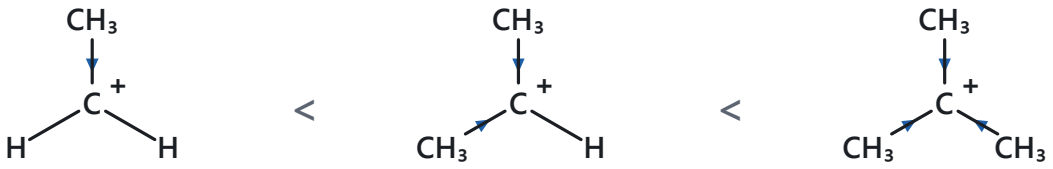

HBr + propene by the main route: H adds to the $CH_2$ end, giving the more stable secondary carbocation and so 2-bromopropane.

Propene is **unsymmetrical**, so the H can add to either carbon of the C=C:

| H adds to | Carbocation formed | Type | Product |
|---|---|---|---|
| the **end** carbon ($CH_2$) | $CH_3–C^+H–CH_3$ | **secondary** (2°): $C^+$ joined to **two** alkyl groups | **2-bromopropane** (**major**) |
| the **middle** carbon (CH) | $CH_3–CH_2–C^+H_2$ | **primary** (1°): $C^+$ joined to **one** alkyl group | 1-bromopropane (minor) |

The arrows are drawn exactly as for ethene. Make sure your intermediate is the **secondary** carbocation if you are asked for the major product.

## Why the secondary carbocation wins




**primary (1°)**
one alkyl group
least stable

**secondary (2°)**
two alkyl groups

**tertiary (3°)**
three alkyl groups
most stable

Each alkyl group pushes electron density towards $C^+$ (small arrows). More alkyl groups spread the positive charge more, so the ion is more stable.

- **Alkyl groups are electron-releasing**: they push electron density towards the positively charged carbon (a positive inductive effect).
- This **spreads out (stabilises) the positive charge**. The more alkyl groups attached to $C^+$, the more stable the carbocation.
- Stability: **tertiary (3°) > secondary (2°) > primary (1°)**.
- The more stable carbocation forms **more easily** (lower activation energy), so most of the product comes from it. That product is the **major product**.

The same reasoning predicts any unsymmetrical case. 2-methylpropene + HBr goes mostly through the **tertiary** carbocation $(CH_3)_3C^+$, giving **2-bromo-2-methylpropane** as the major product.

**KEY IDEA**

Major product = the one formed from the **more stable carbocation intermediate**. Compare the **intermediates**, not the products.

## Evidence for the mechanism

- **A carbocation intermediate is attacked by whatever nucleophile is around.** Ethene + bromine in the presence of **chloride ions** (NaCl solution) gives 1-bromo-2-chloroethane as well as 1,2-dibromoethane, but **no** 1,2-dichloroethane. So the first step always adds a Br atom (the Brδ+ end of $Br_2$), and the second step adds whichever negative ion reaches the $C^+$ first.
- With **bromine water**, water molecules also attack the carbocation, and the main product is **2-bromoethanol** ($BrCH_2CH_2OH$).
- With propene, HBr gives **mostly 2-bromopropane**. That matches the prediction from carbocation stability.

## Unsymmetrical electrophiles

Work out which end is δ+ using electronegativity. The **δ+ atom** adds first (it is the electrophile); the **δ− atom** ends up on the carbocation carbon.

| Electrophile | Dipole | With propene, major product |
|---|---|---|
| HBr | Hδ+–Brδ− | 2-bromopropane, $CH_3CHBrCH_3$ |
| BrCl | **Brδ+**–Clδ− (Cl more electronegative) | 1-bromo-2-chloropropane, $CH_3CHClCH_2Br$ |
| ICl | **Iδ+**–Clδ− | 2-chloro-1-iodopropane, $CH_3CHClCH_2I$ |
| $H_2SO_4$ | Hδ+ on an O–H bond | propyl hydrogensulfate, $CH_3CH(OSO_3H)CH_3$ |

In each case the δ+ atom bonds to the **$CH_2$ end**, so the $C^+$ is on the middle carbon (secondary).

**MARK-SCHEME WORDING**

- HBr mechanism (Jan 2023, 4 marks, six points): **dipole on H–Br**; arrow from the **H–Br bond to Br**; curly arrow from the **C=C to Hδ+**; **correct carbocation**; arrow from a **lone pair on $Br^-$ to $C^+$**; **charge on bromide**. Marks go by how many of these points you include, so leave none out.
- $Br_2$ mechanism (Jan 2019): **induced dipole** on Br–Br; one arrow from C=C to Brδ+; one from the Br–Br bond to (or just beyond) Brδ−; intermediate; lone pair on $Br^-$ with the arrow from it to $C^+$. IGNORE bond angles. Incorrect partial charges are penalised.
- Stability (Jan 2023, 2 marks): "the product is formed from the **secondary carbocation (intermediate)**" and "the **secondary carbocation is more stable** than the primary". Oct 2022: "the **tertiary carbocation** is more stable than the primary"; ALLOW "three methyl groups attached to the positive C". IGNORE Markovnikov's rule.
- BrCl: dipole Brδ+–Clδ− (Jan 2021); ICl: Iδ+ (Jun 2021).

Often asked as: the full mechanism for a named alkene (ethene, propene, 2-methylpropene, cyclohexene, a cyclic alkene); identify the carbocation intermediate; explain why one product is major; spot and correct errors in a given mechanism.

**COMMON MISTAKE**

Arrows starting from an **atom** or from a **charge** instead of from the bond or the lone pair; arrows pointing the wrong way. This is in every examiner report from 2019 to 2024.

No dipole on HBr or $Br_2$, or δ+ on the wrong atom (WCH11 Jan 2019, Oct 2019, Jan 2021 ER).

Writing Brδ− in step 2 instead of a full **$Br^-$** charge with a **lone pair** (WCH11 Oct 2020, Jan 2021, Oct 2022, Jan 2024 ER).

Using ethene when the question names a different alkene (WCH11 Jan 2019, Jan 2023, Oct 2023 ER). Practise with several alkenes.

Using HBr when bromine was asked, or the other way round (WCH11 Jan 2019 ER).

Drawing the primary carbocation when the major product comes from the secondary (WCH11 Jun 2022, Oct 2022 ER).

"2-bromopropane is a secondary carbocation", "secondary halogenoalkanes are more stable", or comparing the stability of the **products**: it is the **intermediate** carbocations you compare (WCH11 Jan 2023, Oct 2023, Oct 2024 ER).

Quoting Markovnikov's rule instead of explaining carbocation stability. The rule is not on the spec and earns nothing by itself (WCH11 Jan 2023 ER).

**⚠ Beyond the spec · not part of the Edexcel IAL specification**

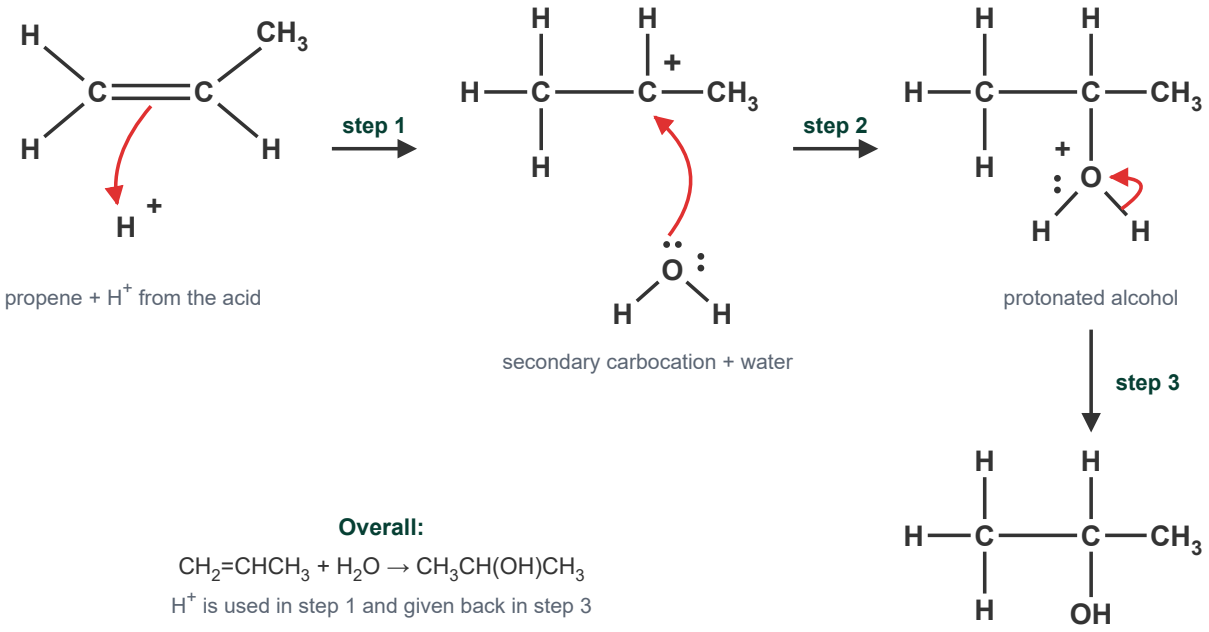
Examined in past papers even though the spec doesn't list it.

**Unsymmetrical electrophiles**: BrCl (WCH11 Jan 2021), ICl (WCH11 Jun 2021), and $H_2SO_4$ adding to an alkene. For $H_2SO_4$: arrow from C=C to the H of an O–H; arrow from that O–H bond to O; carbocation + $HSO_4^-$; lone pair on the $O^-$ of $HSO_4^-$ to $C^+$ (WCH11 Oct 2024, nine-point scheme).

**Acid-catalysed hydration** of propene: arrow from C=C to $H^+$; secondary carbocation; arrow from a **lone pair on the O of $H_2O$** to $C^+$; the O then loses $H^+$, giving **propan-2-ol** and **regenerating $H^+$** (the catalyst) (WCH11 Jan 2021).

How the **π electrons induce a dipole** in $Cl_2$ or $Br_2$ (WCH11 Jan 2025).

Correcting a wrong mechanism: reverse the dipole on $Br_2$; start the arrow at the π bond; give $Br^-$ a lone pair and charge (WCH11 Oct 2019).




**Overall:**

$CH_2{=}CHCH_3 + H_2O \rightarrow CH_3CH(OH)CH_3$

$H^+$ is used in step 1 and given back in step 3

+ $H^+$: the catalyst is regenerated

Three steps: $H^+$ is the electrophile; water attacks the carbocation with a lone pair on oxygen; then the extra $H^+$ is lost and the catalyst is regenerated.

## Acid-catalysed hydration of propene

1. The π bond attacks $H^+$. H adds to the end carbon, giving the more stable secondary carbocation, $CH_3C^+HCH_3$.
2. A lone pair on the oxygen of a water molecule forms a bond to $C^+$. The oxygen now has three bonds and carries the positive charge.
3. An O–H bond breaks, with both electrons going to the oxygen, and $H^+$ is released. The product is propan-2-ol and the catalyst is regenerated.

$$CH_2{=}CHCH_3 + H_2O \xrightarrow{H^+} CH_3CH(OH)CH_3$$

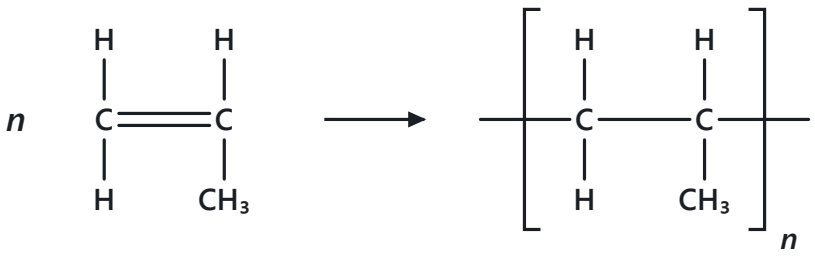
# SPEC 5.7 Addition polymerisation

**WHAT THIS SPEC POINT ASKS**

Describe the addition polymerisation of alkenes. Draw the repeat unit from the monomer, and the monomer from the repeat unit.

## What happens

- **Monomers**: small molecules with a **C=C**.
- Under suitable conditions (high pressure and an initiator, or a catalyst) the **π bond in each monomer breaks** and the monomers join into a long saturated chain: the **polymer**.
- The polymer is the **only product** (atom economy 100%).
- The polymer chain has **no C=C** left. It is saturated and very unreactive, like an alkane.




propene (monomer) poly(propene), shown as its repeat unit

Polymerisation of propene. The repeat unit has two main-chain carbons, the $CH_3$ as a side group, extension bonds through the brackets, and n outside.

## Monomer → repeat unit

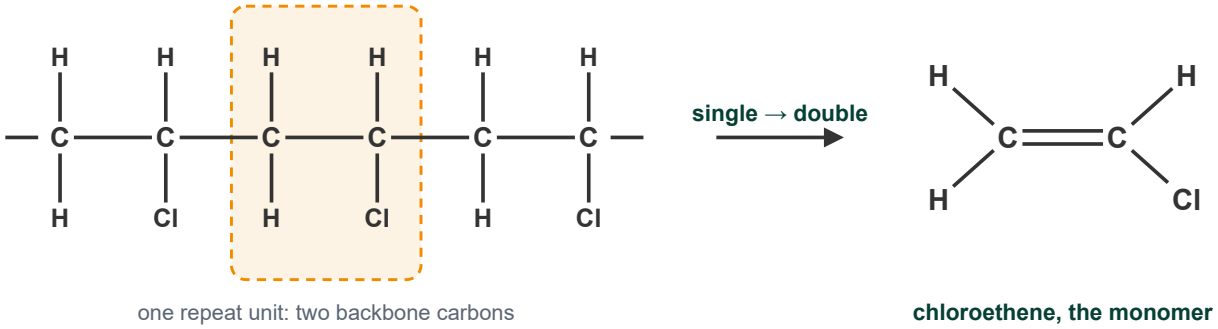
1. Draw the monomer with the **C=C horizontal** and the other four groups **above and below** (not along the line).
2. Change the C=C to a **C–C single bond**.
3. Add an **extension bond** to the left of the first carbon and to the right of the second.
4. Put **square brackets** across the extension bonds and write ***n*** at the bottom right.
5. For the equation, write ***n*** in front of the monomer too.

$n\ CH_2{=}CHCH_3 \rightarrow -[CH_2{-}CH(CH_3)]-_n$

**The repeat unit's main chain is always just two carbons** (from a monomer with one C=C). Everything else hangs off as side groups.

## Repeat unit → monomer

1. Take the **two main-chain carbons** inside the brackets.
2. Remove the extension bonds and the brackets.
3. Put a **C=C** between those two carbons. Keep the side groups.




one repeat unit: two backbone carbons

**chloroethene, the monomer**

Working backwards: box any two neighbouring backbone carbons, keep their four groups, and put the C=C back.

## Common addition polymers

| Monomer | Polymer | Repeat unit |
|---|---|---|
| ethene $CH_2=CH_2$ | poly(ethene) | $–[CH_2–CH_2]–$ |
| propene $CH_2=CHCH_3$ | poly(propene) | $–[CH_2–CH(CH_3)]–$ |
| chloroethene $CH_2=CHCl$ | poly(chloroethene), PVC | $–[CH_2–CHCl]–$ |
| tetrafluoroethene $CF_2=CF_2$ | poly(tetrafluoroethene), PTFE | $–[CF_2–CF_2]–$ |
| phenylethene $CH_2=CHC_6H_5$ | poly(phenylethene), polystyrene | $–[CH_2–CH(C_6H_5)]–$ |
| but-1-ene $CH_2=CHCH_2CH_3$ | poly(but-1-ene) | $–[CH_2–CH(C_2H_5)]–$ |
| but-2-ene $CH_3CH=CHCH_3$ | poly(but-2-ene) | $–[CH(CH_3)–CH(CH_3)]–$ |
| methyl 2-methylpropenoate $CH_2=C(CH_3)COOCH_3$ | Perspex | $–[CH_2–C(CH_3)(COOCH_3)]–$ |

WORKED EXAMPLE

## Worked example: number of repeat units

A sample of poly(propene) has an average $M_r$ of 210 000.

Repeat unit $C_3H_6$: $M_r = (3 \times 12.0) + (6 \times 1.0) = 42.0$

Number of repeat units = 210 000 ÷ 42.0 = **5000**

EXAM TIP

"Draw **two** repeat units" means four main-chain carbons in one continuous chain, with extension bonds at each end. Two separate bracketed units, or one, or three, don't score (WCH11 Jan 2019).

MARK-SCHEME WORDING

- Repeat unit (Jan 2026): "diagram showing **extension bonds**"; ALLOW $CH_3$ written as a group; DO NOT AWARD **missing H** or a skeletal formula.
- Equation (Jun 2019): ***n*** before the monomer **and** after the polymer brackets; "the equation must be balanced".
- Polymer is saturated: **no C=C** in the repeat unit (Oct 2021 ER).
- Addition v condensation (Oct 2020): poly(lactic acid) is a **condensation** polymer that gives off a small molecule (water); propene polymerisation gives **no** other product (addition). ALLOW "higher atom economy".

Often asked as: draw the repeat unit (propene, but-1-ene, chloroethene, tetrafluoroethene); draw two repeat units; deduce the monomer from a polymer section; write the equation with *n*.

COMMON MISTAKE

Three carbons in the main chain (e.g. putting the $CH_3$ of propene into the backbone); leaving out the $CH_3$; keeping the C=C in the repeat unit (WCH11 Jan 2021, Oct 2021 ER).

Drawing poly(ethene) or poly(but-2-ene) when but-1-ene was asked (WCH11 Oct 2021 ER).

Missing *n* on one side; extension bonds that don't pass through the brackets (WCH11 Oct 2020, Jun 2019, Jan 2023 ER).

Saying propene polymerisation "produces hydrogen". Addition polymerisation produces **nothing** else (WCH11 Oct 2020 ER).

Drawing trifluoroethene for tetrafluoroethene: tetra = **four** F (WCH11 Jun 2024).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

**Monomers with two C=C bonds** (dienes). Buta-1,3-diene, $CH_2=CH–CH=CH_2$, polymerises to a repeat unit with **four** main-chain carbons and a C=C in the middle: $–[CH_2–CH=CH–CH_2]–$. Working backwards, a repeat unit with a C=C inside the chain came from a monomer with two C=C bonds (WCH11 Oct 2023, Oct 2022).

$M_r$ of a repeat unit and the number of repeat units in a chain (WCH11 Oct 2023).

Why polymers suit some uses: low density, don't corrode, don't release metal ions into water (WCH11 Jan 2025).

# SPEC 5.8 Dealing with polymer waste

WHAT THIS SPEC POINT ASKS

Understand how chemists reduce the problems of polymer disposal by (i) developing biodegradable polymers and (ii) removing toxic waste gases produced when polymers are burned.

## The problem

- Addition polymers have **strong, non-polar C–C and C–H bonds** and no reactive functional group, so microorganisms' enzymes can't break them down. They are **non-biodegradable** and last for hundreds of years.
- **Landfill**: takes up land; plastics persist; litter harms wildlife.
- **Incineration**: releases energy (which can generate electricity) and takes little space, but makes $CO_2$ and **toxic gases**. Burning **PVC** releases **hydrogen chloride**, HCl. Incomplete burning also makes CO and soot.

## What chemists do

**(i) Develop biodegradable polymers**

- A **biodegradable** polymer is broken down by **microorganisms** (bacteria, fungi) into harmless products such as $CO_2$, water and biomass.
- Examples: **poly(lactic acid), PLA**, made from plant starch (corn, sugar cane). It is a **condensation** polymer with ester links that can be hydrolysed. Starch-based plastics and compostable bags.
- Benefits: less waste to landfill; less litter and harm to wildlife; no need to incinerate; often made from **renewable** plant material rather than crude oil.
- Limits: needs the right conditions (moisture, warmth, microbes) to break down; uses farmland; may still release $CO_2$ or methane as it decomposes.

**(ii) Remove toxic gases from incinerator waste**

- The waste gases are passed through **scrubbers** containing a **base**, which **neutralises** acidic gases such as HCl and $SO_2$.
  - $CaO(s) + 2HCl(g) \rightarrow CaCl_2(s) + H_2O(l)$
  - $Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$
  - $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$
- A **large surface area** (spraying a slurry of the base, or a fine powder) speeds up the reaction.
- HCl is also very soluble, so water sprays help to remove it.

**Other approaches**

- **Recycling**: sorting, melting and remoulding thermoplastics (saves crude oil and energy).
- **Feedstock recycling**: cracking waste polymers back into small molecules (monomers, fuels) to make new chemicals.
- **Energy recovery**: burning to generate electricity, with the gases cleaned.

**MARK-SCHEME WORDING**

- Biodegradable (Oct 2020), three of: takes **less time** to break down; **reduces waste to landfill**; no need to incinerate; reduces litter / harm to wildlife; breaks down to **non-harmful products**; made from **renewable** sources / conserves crude oil. DO NOT AWARD "no waste", "no pollution". "Biodegradable" means **broken down by microorganisms**; "breaks down naturally" alone did not score.
- Incineration (Oct 2021, Jan 2023, Jun 2024): advantages: **energy released** to generate electricity or heat; less space than landfill; pollutants can be trapped. Disadvantages: **toxic / corrosive gases** (name **HCl** from PVC); particulates; not all polymers can be recycled.
- Removing HCl (Jun 2019, Oct 2023): **neutralise with a base**, e.g. **calcium hydroxide** or calcium oxide; a large surface area increases the rate. "Scrubbers" alone isn't enough.
- Chemists limit the problem by (Oct 2022): developing **biodegradable** polymers; **removing toxic waste gases** from incineration; converting waste polymers into useful feedstock.

Often asked as: advantages and disadvantages of incineration and landfill; what biodegradable means and why it helps; how HCl is removed from incinerator gases, with an equation.

**COMMON MISTAKE**

Describing what the public should do (reuse, recycle bins) when the question asks what **chemists** do (WCH11 Oct 2022 ER).

Saying incineration "uses" a lot of energy: it **releases** energy (WCH11 Oct 2021 ER).

Confusing biodegradable with recyclable. Non-biodegradable polymers can be recycled too (WCH11 Oct 2020 ER).

Mentioning $CO_2$ and global warming when the question says not to (WCH11 Oct 2021 ER).

Suggesting a **catalytic converter** to remove HCl (WCH11 Jun 2019 ER). Use a base.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

**Poly(lactic acid)** as a condensation polymer compared with poly(propene) (WCH11 Oct 2020).

**Landfill gas**: anaerobic decay makes methane and $CO_2$ (WCH11 Jun 2022).

## Quick checks

QUICK CHECK

**Which statement about the C=C bond in ethene is correct?**

- It is two σ bonds, so it is twice as strong as a C–C bond.
- It is a σ bond and a π bond, and the π bond is the weaker of the two.
- The π bond lies along the line between the two carbon nuclei.
- The π bond allows free rotation about the C=C bond.

A double bond is one σ bond plus one π bond. The π bond comes from sideways overlap of p orbitals above and below the plane, which is less effective than end-on overlap, so it is weaker (612 is less than 2 × 347 kJ $mol^{-1}$). It prevents rotation rather than allowing it.

QUICK CHECK

**Which compound has E/Z isomers?**

- propene, $CH_2{=}CHCH_3$
- 2-methylpropene, $(CH_3)_2C{=}CH_2$
- 1,1-dichloroethene, $CCl_2{=}CH_2$
- pent-2-ene, $CH_3CH{=}CHCH_2CH_3$

In pent-2-ene each carbon of the C=C carries two different groups (H and $CH_3$; H and $C_2H_5$). In each of the others, one carbon of the C=C carries two identical groups.

QUICK CHECK

**In CHBr=CHCl, the Br and the Cl are on opposite sides of the double bond. What is the isomer?**

- E-1-bromo-2-chloroethene
- Z-1-bromo-2-chloroethene
- It has no E/Z isomers.
- It cannot be named, because Br and Cl are different groups.

On each carbon the halogen beats H (35 or 17 against 1), so Br and Cl are the priority groups. Opposite sides means E. Cis–trans would struggle here, but E/Z works for any groups.

QUICK CHECK

**What is the major product when hydrogen bromide reacts with 2-methylpropene, $(CH_3)_2C=CH_2$?**

- 1-bromo-2-methylpropane
- 2-bromo-2-methylpropane
- 1,2-dibromo-2-methylpropane
- 2-methylpropan-2-ol

H adding to the $CH_2$ carbon leaves a tertiary carbocation, $(CH_3)_3C^+$, stabilised by three electron-releasing methyl groups. $Br^-$ adds to that carbon. The other route goes through a much less stable primary carbocation and gives the minor product, 1-bromo-2-methylpropane.

QUICK CHECK

**In the mechanism of the reaction between ethene and bromine, where does the first curly arrow start?**

- At a lone pair on the $Br^{\delta-}$ atom
- At the Br–Br bond
- At the C=C double bond (the π bond)
- At a C–H bond

The π electrons are the electron source: they move to form a bond to $Br^{\delta+}$. The arrow from the Br–Br bond is the second arrow, showing heterolytic fission. A lone pair on $Br^-$ is used only in the final step, when the bromide ion attacks the carbocation.

QUICK CHECK

**Which reagent and observation show that hex-1-ene is unsaturated?**

- Bromine water turns from colourless to orange.
- Acidified potassium manganate(VII) turns from colourless to purple.
- Universal indicator turns red.
- Bromine water turns from orange to colourless.

Bromine adds across the C=C, so the orange colour of the bromine disappears. Manganate(VII) also works, but it starts purple and is decolourised, not the other way round.

QUICK CHECK

**Which is the repeat unit of the polymer made from chloroethene, $CH_2$=CHCl?**

- –[$CH_2$–CHCl–$CH_2$–CHCl]$_n$–
- –[$CH_2$–CHCl]$_n$–
- –[CH=CCl]$_n$–
- –[$CH_2$–$CH_2$–Cl]$_n$–

The repeat unit is just two backbone carbons with the original four atoms (three H, one Cl) still attached, and the double bond has become single. The first option shows two repeat units, the third keeps a double bond, and the last puts the Cl in the chain.

QUICK CHECK

**Why can burning waste PVC in an incinerator be a problem, and how is it dealt with?**

- It releases chlorine gas, which is removed by passing the gases through water.
- It releases hydrogen chloride, which is removed by reacting it with a base such as calcium oxide.
- It releases sulfur dioxide, which is removed with a catalytic converter.
- It does not burn, so it must go to landfill.

PVC contains chlorine, which ends up as acidic, toxic HCl gas when it burns. Passing the waste gases through a base neutralises it: $CaO + 2HCl \rightarrow CaCl_2 + H_2O$.